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ADK vs RAG: How to Choose the Right AI Stack

Use a simple test — should the AI act or should it recall — to decide between an agent architecture, a retrieval pipeline, or both.

For developers and technical product people deciding whether to build an AI agent, a retrieval system, or a combination of the two · 27 July 2026

Every section, key point, term and question from the full lesson is here. The extended explanations and the additional examples are not.

[Watch the source video](#)

[Read the full lesson](#)

[Read the highlights](#)

BEFORE YOU START

- Basic understanding of what a large language model (LLM) does when it generates text
- Familiarity with the idea of calling a function or API from code

Overview

Every AI system design eventually runs into the same fork: should the system perform actions, or should it answer questions grounded in a body of knowledge? Agent architectures (often built with an Agent Development Kit, or ADK) are optimized for the first job — following instructions, calling tools, and executing multi-step workflows. Retrieval-Augmented Generation (RAG) is optimized for the second — grounding answers in documents so the model isn't relying on guesswork baked into its training data.

These aren't competing technologies so much as answers to different questions. An agent framework gives a language model hands: it can call a function, follow a decision tree, and keep doing that reliably every time it's asked. A retrieval pipeline gives a language model a library card: before it answers, it looks something up, so its response reflects your documents instead of whatever it happened to memorize during training.

Most production systems that look sophisticated from the outside are not choosing one over the other — they're combining them. An agent orchestrates the steps of a task, and one or more of those steps calls a retrieval pipeline to fetch grounded facts before continuing. Understanding when each piece is doing the work, and why, is what makes that combination deliberate instead of accidental.

KEY TAKEAWAYS

- ✓ Agent architectures (ADK-style) are for AI that needs to act: follow steps, call tools, and produce consistent, repeatable behavior.
- ✓ RAG is for AI that needs to know: it retrieves relevant text from your documents before generating an answer, so the answer is grounded in that source instead of the model's internal training.
- ✓ The fastest way to choose is to ask one question: does this system need to act, or does it need to recall?
- ✓ ADK suits multi-step workflows, tool use, and tasks where the value comes from reasoning and repeatable logic, not memory.
- ✓ RAG suits domains where the source of truth is a document set that is large, detailed, or changes often — retrieval keeps answers current without retraining a model.
- ✓ Hybrid systems are the norm in real deployments: the agent layer handles task flow and decisions, while a retrieval layer supplies it with accurate, up-to-date facts.
- ✓ A retrieval step and a tool-calling step are not mutually exclusive — retrieval can itself be exposed to an agent as just another tool it decides to call.

01 The Hardware Store Analogy: Two Kinds of AI Systems

0:00

A hardware store has a tool aisle for doing work and a reference aisle for knowing how to do it correctly — AI systems split the same way.

KEY POINTS

- Tools act on the world; reference material informs decisions but doesn't act on its own.
- Neither approach is inherently more advanced — they answer different questions.
- The analogy maps directly onto two AI architecture styles covered next: ADK (acts) and RAG (knows).

Fixing a bookshelf

Building a bookshelf needs a drill (a tool, acting) and possibly a joinery diagram (a reference, informing). You wouldn't try to drive a screw with the diagram, and you wouldn't expect the drill to tell you which screw to use. Most real carpentry projects, like most real AI systems, need both.

ADK-style systems give an AI model the ability to reason step by step and call tools to complete multi-step tasks predictably.

KEY POINTS

- ADK-style agents run in a loop: reason about the next step, call a tool, observe the result, reason again.
- Core traits: step-by-step reasoning, tool use, explicit rules/logic, and repeatable behavior across runs.
- Repeatability is what makes agent behavior testable — the same input tends to trigger the same decision path.
- The agent's value is in reasoning and acting, not in recalling facts it was never given.

A minimal tool-calling agent loop

This is the mechanical skeleton every ADK-style agent shares: give the model a set of callable tools, let it decide whether to call one or answer directly, feed the tool's result back in, and repeat until it has enough information to respond. The 'llm' and 'tools' objects stand in for a real model client and real functions.

```
def run_agent(user_request, tools, llm, max_steps=6):
    """A minimal ReAct-style agent loop: reason, act, observe, repeat."""
    messages = [{"role": "user", "content": user_request}]

    for step in range(max_steps):
        response = llm.chat(messages, tools=tools)

        if response.tool_call is None:
            # Model decided it has enough information to answer directly
            return response.content

        # Execute the tool the model asked for
        tool_name = response.tool_call.name
        tool_args = response.tool_call.arguments
        result = tools[tool_name](**tool_args)

        # Feed the observation back so the model can reason further
        messages.append({"role": "assistant", "tool_call": response.tool_call})
        messages.append({"role": "tool", "name": tool_name, "content": str(result)})

    return "Max steps reached without a final answer."
```

03 RAG: Designing AI That Retrieves and Knows

2:06

RAG connects a model to your documents and retrieves the relevant pieces before it

answers, trading the model's internal guesswork for grounded, sourced text.

KEY POINTS

- RAG retrieves relevant document chunks before generating a response, instead of answering from memorized training data.
- Documents are chunked and embedded into vectors; a vector database finds chunks whose meaning is closest to the question.
- Retrieved chunks are inserted into the prompt as context that the model is told to rely on.
- Updating the underlying documents changes future answers immediately — no retraining required.

A minimal RAG answer function

This function embeds the incoming question, retrieves the top-k most relevant chunks from a vector store, builds a context block from them, and instructs the model to answer only from that context. If nothing relevant was retrieved, the model is told to say so rather than fill the gap with a guess.

```
def answer_with_rag(question, vector_store, llm, top_k=4):
    """Retrieve relevant document chunks, then generate a grounded answer."""
    query_embedding = embed(question)
    chunks = vector_store.search(query_embedding, top_k=top_k)

    context = "\n\n".join(f"[{c.source}] {c.text}" for c in chunks)
    prompt = (
        "Answer using ONLY the context below. "
        "If the answer isn't there, say so.\n\n"
        f"Context:\n{context}\n\nQuestion: {question}"
    )
    return llm.complete(prompt)
```

04 Choosing ADK: When Your AI Needs To Do Things

2:51

The core test is 'act or recall?' — ADK says 'do something for me,' and it fits anywhere

the value comes from reasoning through steps rather than looking something up.

KEY POINTS

- The one-question test: does the AI need to act (ADK) or recall (RAG)?
- ADK fits multi-step workflows, content drafting/transformation, IT/HR assistance, and triage or routing tasks.
- Choose ADK when the value comes from reasoning through a decision, not from looking up a fact.
- Consistent, repeatable logic makes ADK-style systems easier to evaluate than free-form generation.

IT support triage as a procedural task

A request like 'my VPN keeps disconnecting' needs to be classified (network vs. account issue), routed to the right queue, and possibly have a ticket created — the same three steps every time, regardless of which specific document might describe VPN troubleshooting. That repeatable sequence is exactly what an ADK-style agent is built to execute.

05 Choosing RAG: When Your Data Is the Source of Truth

3:37

Choose RAG when accuracy has to come directly from documents, especially where content is high-volume, detailed, or constantly changing.

KEY POINTS

- Use RAG when accuracy must trace back to a document, not to the model's internal guesswork.
- Best-fit content: PDFs, policies, regulations, technical docs, product manuals, long-form knowledge bases.
- RAG handles information too large or detailed for a person (or a fixed prompt) to hold in full.
- It also handles widely varying question phrasing, since retrieval matches on meaning rather than a fixed script.

Legal document lookup

A question like 'does our vendor contract with Acme allow termination for convenience' can't be answered from a model's training — it depends on the specific contract text. A RAG system retrieves the relevant clause from the actual contract PDF and answers from that clause, rather than producing a plausible-sounding but unverified guess.

06 Hybrid Systems: Reasoning Paired With Domain Knowledge

4:20

Most real systems use both: an ADK-style agent handles the task flow and decisions, while a RAG component supplies it with grounded facts along the way.

KEY POINTS

- Hybrid = ADK controls the task flow and decisions; RAG supplies grounded facts at the steps that need them.
- Content generation / high reasoning / low retrieval -> ADK. Answers dependent entirely on documents -> RAG.
- Domain-expert co-pilots (legal, engineering, healthcare) typically need both deep retrieval and complex reasoning.
- The cleanest integration exposes retrieval to the agent as one more callable tool, not a separate parallel system.

A support agent with retrieval as a tool

The agent loop from the ADK section is reused unchanged. One of its tools, 'retrieve_docs', is simply the RAG function from the previous section. The agent decides on its own when a step needs a grounded fact and calls that tool, then continues reasoning and acting with whatever it learned.

```
tools = {
    "retrieve_docs": lambda query: answer_with_rag(query, vector_store, llm),
    "create_ticket": create_ticket,
    "send_email": send_email,
}

run_agent(
    "A customer is asking about our refund policy for damaged items. "
    "Look it up and draft a reply.",
    tools=tools,
    llm=llm,
)
```

07 A Practical Decision Checklist

5:42

Back to the hardware store: use the tool aisle to build, the reference aisle to make sure the build is correct, and pick based on whether the task is to act, to know, or both.

KEY POINTS

- ADK = tool aisle (acts, builds); RAG = reference aisle (informs, grounds).
- Two questions decide the architecture: does it need to act? Does it need current, document-grounded knowledge?
- Four possible outcomes: ADK only, RAG only, hybrid, or a plain model call with no extra architecture.
- Deciding this before picking a framework prevents over- or under-building the system.

A two-question decision function

A direct translation of the hardware-store heuristic into code. It's deliberately simple — the point isn't the code, it's that the two questions are enough to place almost any system into the right quadrant before any framework gets chosen.

```
def choose_architecture(needs_action, needs_current_docs):
    if needs_action and needs_current_docs:
        return "Hybrid: ADK orchestrates the task, RAG supplies grounded facts"
    if needs_action:
        return "ADK: a procedural, tool-using agent"
    if needs_current_docs:
        return "RAG: retrieval over your documents"
    return "A plain model call may be enough"

# Examples from the lesson
print(choose_architecture(needs_action=True, needs_current_docs=False)) # IT triage
print(choose_architecture(needs_action=False, needs_current_docs=True)) # legal lookup
print(choose_architecture(needs_action=True, needs_current_docs=True)) # legal co-
pilot
```

Glossary

ADK (Agent Development Kit)

A term for agent-style architectures built around action and reasoning — the AI follows steps, calls tools, and produces consistent, repeatable behavior.

RAG (Retrieval-Augmented Generation)

An architecture where a model retrieves relevant text from a document store before generating its answer, so the answer is grounded in that source material.

Agent

An AI system that runs in a loop, deciding on each turn whether to call a tool or produce a final answer, rather than generating one response in isolation.

Tool use / function calling

The mechanism by which a model can invoke an external function or API mid-conversation and receive its result back as new information to reason over.

Embedding

A numeric vector representation of a piece of text, positioned so that text with similar meaning ends up close together in vector space.

Vector database

A database optimized for storing embeddings and finding the ones closest in meaning to a given query vector.

Grounding

Basing a model's answer on retrieved source material rather than on patterns memorized during training, so the answer can be traced back to a specific document.

ReAct pattern

An agent loop of reason, then act, then observe the result, then reason again — the mechanical basis of most tool-calling agents.

Triage

Classifying and routing an incoming request (a ticket, a message) to the right handler or process based on rules or logic.

Hybrid system

An architecture where an agent (ADK-style) controls the task flow and calls a retrieval pipeline (RAG) as one of its tools whenever a step needs grounded information.

Check yourself

Answer first, then open each one.

A support bot answers questions about a 200-page policy manual that gets revised every month. Why would a RAG system stay accurate after a revision while a plain fine-tuned model would not?

RAG retrieves from the live document at answer time, so an edit to the source document is reflected in the very next query. A fine-tuned model bakes the old text into its weights at training time; it keeps answering from the stale version until it's retrained, which is slower and more expensive than just updating a document.

Why is an ADK-style agent easier to evaluate than a single free-form model generation?

Because the agent follows explicit, repeatable steps (reason, call a tool, observe, repeat), the same input tends to produce the same decision path each run. That repeatability lets you inspect and test individual steps, whereas a single open-ended generation gives you no intermediate points to check.

In the hybrid support-agent example, retrieval is exposed to the agent as just another tool rather than as a separate pipeline running alongside it. Why does that design choice matter?

Keeping retrieval inside the agent's own tool set means one control loop decides when grounded information is actually needed for the current step, instead of always running retrieval whether or not the step calls for it. It also means the retrieved result flows back into the agent's next reasoning step naturally, the same way any other tool result would.

A multi-step onboarding workflow (create account, assign role, send welcome email) doesn't reference any documents. Why would adding a RAG layer here likely be wasted effort?

The task's value comes entirely from executing a fixed sequence correctly, not from looking anything up — there's no changing, document-based fact the workflow needs to consult. RAG solves a knowledge problem, and this task doesn't have one.

Why does 'does the answer change when I update a document' work as a quick test for whether a system needs RAG?

If updating a document should change future answers, the system's correctness depends on live document content, not on task logic — which is exactly the situation retrieval is built for. If the answer wouldn't change no matter what the document said, the system is really about executing steps, which points to an agent architecture instead.

Where to go next

- Build a minimal ReAct-style tool-calling loop in Python against a real LLM API and give it two or three simple tools (a calculator, a date lookup, a mock ticket creator).
- Implement a small RAG pipeline end to end: chunk a handful of PDFs, embed them with an open-source embedding model, store the vectors in a local vector database (e.g. Chroma or FAISS), and query it.

- Take one real workflow you're considering automating and run it through the two-question checklist (needs_action, needs_current_docs) before choosing a framework.
 - Look up Google's Agent Development Kit documentation to see a concrete, production implementation of the agent loop pattern described in this lesson.
 - Read about the ReAct paper (Yao et al., 2022) for the original formulation of the reason-act-observe loop underlying most modern tool-calling agents.
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Explanations are AI-generated. Check anything you intend to rely on.